



Module 7 Day 4: Activity 3, Investigating User Space and Services

Imagine you are a system administrator responsible for securing a company's network. Recently, your team has received reports of unusual activity on several user accounts. As part of your role, you need to investigate these reports, identify any suspicious activity, and propose mitigation steps to prevent future incidents. In this activity, you will simulate the investigation process by analyzing user permissions, inspecting common malware hiding spots, and examining processes and services.

You will begin by evaluating user accounts to ensure that permissions align with the principle of least privilege. Next, you will delve into critical system locations—such as **Tasks**, **Temp**, and **AppData**—to uncover potential malicious files. By examining processes and services, you will refine your ability to spot anomalies that might otherwise go unnoticed.

This hands-on exercise is designed to replicate real-world scenarios where vigilance and a methodical approach are vital. Your findings will directly contribute to enhancing the security posture of your simulated environment, preparing you for challenges you may face in professional settings.

Instructions:

1. On a lab or personal machine, perform the following tasks:
 - **Review User Permissions:** Open **Computer Management** and navigate to Local Users and Groups. Identify accounts with elevated permissions and propose adjustments. Provide reasoning for changes in privileges.
 - **Explore Malware Hiding Spots:** Using **File Explorer**, navigate to:
 - **C:\Windows\Tasks**
 - **C:\Windows\Temp**
 - **C:\Users\<username>\AppData**
 - Document any suspicious files by noting their names, creation dates, and any unusual properties.
 - Use tools like **PowerShell** to search for hidden or system files within these directories.
 - **Investigate Processes:** Open **Task Manager** and sort processes by CPU usage. Note the top 5 highest-consuming processes. Cross-reference their

names and details online to confirm legitimacy. Use tools like **Process Explorer** for deeper insights.

- **Analyze Windows Services:** Open the **Services** tool and identify services with unclear or unfamiliar names. Use a search engine to validate their purpose and origin. For any suspicious services, document the service name, description, and executable path.

2. Compile a report summarizing:

- Changes proposed to user privileges and their justification.
- Suspicious files, processes, or services identified and steps taken for mitigation.
- Insights gained during the activity and any challenges encountered.